SylvanSync™

CONTINGENCY LESSONS

Mathematics for SylvanSync™

Sylvan Learning[sm]

ANSWER KEY

To be used only in case of emergency, with the approval of Technology Services.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Answers

Part 1

Step 1: 7; 9; students are there in all
Step 2: add
Step 3: 9; 16; 16
Step 4: Check student's drawing.

Part 2

1. It was sunny 9 days in the last two weeks.
   a. It was sunny 3 days last week and 6 days this week. How many days has it been sunny in the last two weeks?
   b. Add the number of sunny days last week and this week.
   c. 3 + 6 = 9
   d. Check student's work.

2. Dagney has 24 guppies now.
   a. The first step is to determine what information you know and what is being asked.
   b. Add the number of guppies together.
   c. 13 + 11 = 24
   d. Check student's work.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Part 3

1. There are 8 blocks in all. Check student's work.
2. There are 12 birds all together. Check student's work.
3. There are 13 crayons all together. Check student's work.
4. Gretchen spent 9 dollars. Check student's work.

Mathematics for SylvanSync™
Grades K–2
Problem-Solving Strategies

Answers
Part 1

1. The next two shapes are a yellow square and a red triangle. Check student's work.
2. The desks are in the following order: yellow, orange, blue, red; or red, blue, orange, yellow. Check student's work.
3. Tammy has 21 cupcakes. Check student's work.
4. Maggie's dad has one penny, three nickels, and one dime. Check student's work.
5. Each friend got 4 cookies. Check student's work.

Mathematics for SylvanSync™
Grades K–2
Problem-Solving Strategies

Part 2

1. Mark has run the most miles.

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Joe	3	off	6	off	9	off	12
Mark	2	4	6	8	10	12	14

2. There are 8 combinations of outfits that Erika could make.

3. There are four combinations that Jordan can select.

Shirt	Shorts	Hat
Blue	Green	Black
Blue	Green	Purple
Blue	Yellow	Black
Blue	Yellow	Purple
Orange	Green	Black
Orange	Green	Purple
Orange	Yellow	Black
Orange	Yellow	Purple

baked potatoes and pork
baked potatoes and chicken
mashed potatoes and pork
mashed potatoes and chicken

Mathematics for SylvanSync™
Grades K–2
Problem-Solving Strategies

Part 3

1. Jane has 12 pieces of gum.
   Rose has 7.
   Jacob has 3 more than Rose does. 7 + 3 = 10
   Jacob has 10 pieces. Jane has 2 more than Jacob does.
   10 + 2 = 12

2. Lisa left at 1:25.
   Ashley left at 1:00.
   Cassie left 30 minutes after Ashley; she left at 1:30.
   Lisa left 5 minutes before Cassie.

3. Misty started with $20.00.
   Misty had $7.00 after spending $1.00, $3.00, $4.00, and $5.00.
   $7.00 + $1.00 + $3.00 + $4.00 + $5.00 = $20.00

Mathematics for SylvanSync™
Grades K–2
Flash Cards

Answers

Student responses may vary.

Mathematics for SylvanSync™
Grades K–2
Counting

Answers
Part 1
1. 3
2. 5
3. 10

Part 2
1. 6
2. 8

Part 3
1. Student responses may vary.
2. Student responses may vary.

Part 4
1. 9 shaded squares
2. 7 shaded squares
3. 4 shaded squares

Mathematics for SylvanSync™
Grades K–2
Sorting

Answers

Student responses may vary.

Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle

Answers

Part 1

1. yellow; odd
2. It is a purple number, which is an even number.
3. It is a purple number, which is an even number.
4. It is a yellow number, which is an odd number.

Part 2

1. Yes; the rows repeat from the middle moving out to the left and to the right.
2. The number is the sum of the two numbers above it.
3. The number below is the sum of the two numbers above it.
4. 1, 9, 36, 84, 126, 126, 84, 36, 9, 1
   1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1
5. Check student's work.

Part 3

1. even
2. even
3. odd

Part 4

1. When folded in half the same colors fall on top of each other.
2. Student responses may vary.

Mathematics for SylvanSync™
Grades K–2
Comparing Sets

Answers

Part 1

1. 9 counters
2. more than 9 counters
3. fewer than 9 squares
4. Check student's work.

Part 2

1. circle second row
2. circle top row
3. circle second row
4. circle top row
5. circle second row

Part 3

Student responses may vary.

Mathematics for SylvanSync™
Grades K–2
Introducing Base 10

Answers

Part 1

Check the student’s work.

Part 2

Check the student’s work.

Part 3

1. 74
2. 190
3. 88
4. 400
5. 62
6. 55
7. 127
8. 263

Mathematics for SylvanSync™
Grades K–2
Shapes in the Plane

Answers
Part 1
1. b
2. d
3. e
4. c
5. a

Part 2
1. Check the student’s work.
2. Check the student’s work.
3. Check the student’s work.
4. Check the student’s work.

Part 3
1. Check the student’s work.
2. Check the student’s work.
3. Check the student’s work.
4. Check the student’s work.
5. Check the student’s work.
6. Check the student’s work.
7. Check the student’s work.

Part 4
Check the student’s work.

Part 5
Check the student’s work.

Mathematics for SylvanSync™
Grades K–2
Graphs and Graphing

Answers
Part 1
1. Week 4
2. Week 3
3. 13 books
4. 26 books
5. Week 3 because the fewest number of books were read during that week.

Part 2
1. bowling
2. 3 more
3. 16
4. There are the same number of votes.

Part 3
Check the student's pictograph.

Part 4
1. Check the student's pictograph.
2. Student responses may vary.
3. Check the student's pictograph.

Mathematics for SylvanSync™
Grades K–2
Hundreds Chart

Answers
Part 1
Student responses may vary.

Part 2
1. The ones digits are 1s. The numbers increase by 10.
2. The ones digits are 2s. The numbers increase by 10.
3. 34; 10; yes
4. difference of 10

Part 3
1. Check the student's work.
2. Check the student's work.
3. They are odd numbers.
4. They are even numbers.

Part 4
1. It takes 10 moves.
2. Yes, move down one row.
3. It is increasing or decreasing by 10.
4. Student responses may vary. Acceptable answers include: move down one and to the right two.
5. Student responses may vary. Acceptable answers include: move down three and to the right two.

Mathematics for SylvanSync™
Grades K–2
Evens and Odds

Answers

Part 1

Check the grouping of blocks for each answer.

1. even
2. even
3. odd
4. odd
5. even
6. odd
7. odd
8. even
9. even

Part 2

1. even
2. odd
3. odd
4. even
5. even

Part 3

Check the underlined digit for each answer.

1. odd
2. odd
3. even
4. odd
5. even

Part 4

Student responses for Exercises 20 through 24 may vary.

Mathematics for SylvanSync™
Grades 3–5
The Problem-Solving Process

Answers
Part 1
1. Plan, Solve
2. Plan
3. Look back
4. Solve

Part 2
1. Understand
2. Look back
3. Solve

Part 3
The steps are:
- Step 1: Understand – C
- Step 2: Plan – D
- Step 3: Solve – A
- Step 4: Look back – B

Mathematics for SylvanSync™
Grades 3–5
The Problem-Solving Process

Part 4

1. Student responses may vary. Acceptable answers include:
   Step 1: Understand
   - There are 32 computers to give away. Each school will receive 4 computers. How many schools will receive computers?

   Step 2: Plan
   - Divide the total number of computers by the number each school will receive.

   Step 3: Solve
   - $32 \div 4 = 8$
   - Eight schools will receive computers.

   Step 4: Look back
   - The answer is reasonable. $8 \times 4 = 32$

2. Student responses may vary. Acceptable answers include:
   - Suggested Solution: Melinda has $50.00 and spends $22.00 and $12.00. How much money does she have left?

   Step 2: PLAN
   - Add the purchases together and subtract from $50.00.

   Step 3: SOLVE
   - $22.00 + $12.00 = $34.00
   - $50.00 – $34.00 = $16.00.
   - She has $16.00 left.

   Step 4:
   - The answer is reasonable. $22.00 + $12.00 + 16.00 = $50.00

Mathematics for SylvanSync™
Grades 3–5
The Problem-Solving Process

3. Student responses may vary. Acceptable answers include:
   Step 1: Understand
   - Bo earns $8 per hour of cleaning and $18 per lawn. He cleans for 4 hours and mows 3 lawns. How much did he earn?

   Step 2: Plan
   - Multiply the number of hours of cleaning by the wage and the number of lawns by the wage. Add the two products.

   Step 3: Solve
   - $\$8 \times 4 = \$32$
   - $\$18 \times 3 = \$54$
   - $\$32 + \$54 = \$86$
   - He earned $86 this week.

   Step 4: Look back
   - The answer is reasonable.

4. Student responses may vary. Acceptable answers include:
   Step 1: Understand
   - There are 597 pairs of socks. The sale is on three pairs. Can 597 pairs of socks be sold in groups of 3?

   Step 2: Plan
   - Divide 597 by 3.

   Step 3: Solve
   - $597 \div 3 = 199$
   - All the socks could be sold in groups of three if 199 pairs of socks were sold.

   Step 4: Look back
   - The answer is reasonable. $199 \times 3 = 597$

5. Student responses may vary. Acceptable answers include:
   Step 1: Understand
   - Apples cost $1.80 for four.
   - How much would 10 apples cost?

   Step 2: Plan
   - Divide $1.80 by 4 in order to determine the cost of one apple.
   - Take that amount and multiply by 10.

   Step 3: Solve
   - $\$1.80 \div 4 = \$0.80$
   - $\$0.45 \times 10 = \$4.50$
   - It would cost $4.50 for 10 apples.

   Step 4: Look back
   - The answer is reasonable.

Mathematics for SylvanSync™
Grades 3–5
Problem-Solving Strategies

Answers
Part 1
1. The Amur leopards have the fewest animals remaining.
2. There are 30 fewer remaining red wolves.
3. The total population of Amur leopards, Siberian tigers, and Florida panthers is 620.
4. The total double population of the seven species is 3800.
5. Student responses may vary. Acceptable answers include: The table helps to organize the information and makes it easier to select the needed numbers.

Part 2
Students' strategies may vary. Acceptable answers include:

1. make a model; draw a picture
   There are 4 kilometers between the water tables.
2. draw a picture
   Misty has traveled 95 miles.
3. make a table; find a pattern
   Bryson can expect to score 20 points in the 10th game if the pattern continues.
4. draw a picture; guess and check
   Marquis has a ten dollar bill, a five dollar bill, a one dollar bill, a quarter, two dimes, and six pennies.
5. work backwards
   Sheila started running at 4:18 pm.

Part 3
1. Student responses for strategies may vary. Acceptable answers include: make a model; make a table or list; find a pattern.
   They will work for 12 hours in the sixth month if they continue to follow the pattern.
2. Student responses for strategies may vary. Acceptable answers include: make a model; make a table or list; find a pattern
   Natalie will score 15 points in game six if she continues to follow the same pattern.

Mathematics for SylvanSync™
Grades 3–5
Flash Cards

Answers
Student responses may vary.

Mathematics for SylvanSync™
Grades 3–5
Pascal's Triangle

Answers
Part 1
1. The number below is the sum of the two numbers above it.
2. Using two numbers above, find the sum and place it in the space below.
3. Check the student's worksheet.

Part 2
1. Check the student's worksheet.
2. Check the student's worksheet.

Part 3
1. They contain both colors because 36 is a multiple of both 3 and 4.
2. yes
3. Student responses may vary.

Part 4
1. Check the student's explanation.
2. Check the student's explanation and sketch.

Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Answers
Part 1
Student models may vary. Check each student's work for accuracy.
1. Ten buttons drawn with 7 shaded.
2. Three models divided into tenths are drawn with 2 whole models shaded and 4 of the 10 parts of the last model shaded.
3. One model divided into tenths with nine-tenths of the model purple and one-tenth green.
4. Four models divided into tenths are drawn with 3 whole models shaded and 5 of the 10 parts of the last model shaded. Alternately the student may have recognized that five-tenths is equal to one-half and divided the models into two parts.

Part 2
1. 0.56
2. 2.11
3. 0.06
4. 0.99
5. 1.78

Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Part 3

1. fractions:
   a. $\frac{2}{10}$
   b. $\frac{1}{3}$
   c. $\frac{1}{8}$
   d. $\frac{3}{4}$
   e. $\frac{3}{5}$
   f. $\frac{7}{8}$
   g. $\frac{1}{6}$
2. decimals:
   a. 0.25
   b. $0.08\overline{3}$
   c. 0.5
   d. 0.70
   e. $0.6\overline{6}$
   f. 0.625

Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Part 4

	Answer Given as a Fraction	Answer Given as a Decimal
1. $\frac{1}{10} + 0.2 =$	$\frac{3}{10}$	0.3
2. $0.5 + \frac{1}{4} =$	$\frac{3}{4}$	0.75
3. $0.08\overline{3} + \frac{11}{12} =$	1	1
4. $\frac{1}{8} + 0.25 =$	$\frac{3}{8}$	0.375

Mathematics for SylvanSync™
Grades 3–5
Understanding Fractions

Answers
Part 1
1. Check the student's work.
2. Student responses may vary.
3. Check the student's explanation; $\frac{3}{4}$

Part 2
1. Student responses may vary.
2. $\frac{6}{8}$; $\frac{2}{3}$; form
3. $\frac{5}{6}$
4. Check the student's explanation; $\frac{2}{3}$

Part 3
1. equivalent fractions:
   a. 2
   b. 3; 2
   c. 3
   d. 4; 2
2. sums:
   a. $\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$
   b. $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$
   c. $\frac{4}{10} + \frac{3}{10} = \frac{7}{10}$

Mathematics for SylvanSync™
Grades 3–5
Understanding Fractions

Part 4

1. $\frac{4}{6}; \frac{2}{3}$
2. $\frac{2}{12}; \frac{1}{6}$
3. 1
4. $\frac{2}{5}$
5. $\frac{5}{12}$
6. $\frac{1}{4}$

Mathematics for SylvanSync™
Grades 3–5
Plane Shapes

Answers
Part 1
1. Check the student’s work.
2. Check student’s work.
3. Check student’s work.
4. Check student’s work.
5. Check student’s work.

Part 2
Check the work on the Geoboards.
1. 8 square units
2. perimeter= 18 units; area = 14 square units
3. area = 15 square units
4. area = 28 square units

Part 3
Check the work on the Geoboards.
1. 15 square units
2. 31 square units

Part 4
Student responses may vary. Check the work on the Geoboards.

Mathematics for SylvanSync™
Grades 3–5
Shapes in Space

Answers

Part 1

1. sphere
2. cylinder
3. hemisphere
4. hexagonal prism
5. triangular prism
6. octagonal prism
7. square pyramid

Part 2

Student responses may vary. Check the student’s explanations.

Part 3

Check the student’s table.

Mathematics for SylvanSync™
Grades 3–5
Wrap-ups: Concept Practice

Answers
Part 1
1. There are four possible outcomes: yellow, orange, purple, and pink.
2. The 52 cards in the deck are the 52 possible outcomes.
3. There are two possible outcomes: heads; tails.
4. The 10 numbers are the 10 possible outcomes.

Part 2
1. marbles:
   a. $\frac{5}{30}$ or $\frac{1}{6}$
   b. $\frac{15}{30}$ or $\frac{1}{2}$
   c. $\frac{0}{30}$ or 0; Since there are no white marbles in the bag it is impossible to select a white marble.
2. letters:
   a. $\frac{5}{26}$
   b. $\frac{21}{26}$
   c. $\frac{2}{26}$ or $\frac{1}{13}$

Part 3
Student responses may vary. Check the student's charts.

Part 4
Student responses may vary. Check the student's charts.

Part 5
Student responses may vary. Check the student's answers and chart.

Mathematics for SylvanSync™
Grades 6–8
The Problem-Solving Process

Answers

Part 1

1. Step 3: Solve the problem.
2. Step 2: Develop a plan.
3. Step 4: Look back.
4. Step 1: Understand the problem.

Part 2

1. a. Student responses may include: Brielle bought a sandwich for $4.25, a drink for $1.50, and fruit salad for $2.75. She gave the clerk $10.00.
   b. Student responses may include: How much change did she receive?

2. Student responses may include: Find the total amount that Brielle spent and subtract it from $10.00.
3. $1.50
4. Student responses may vary.

Part 3

1. Tyler ran an average of 33 minutes each day.
   Check student's process and solution.

2. Hannah will need to babysit 6 hours to earn enough money to buy the sweater and pants.
   Check student's process and solution.

3. Irina has 54 miles to travel.
   Check student's process and solution.

4. The area of the remaining sheet is 55 square feet.
   Check student's process and solution.

Mathematics for SylvanSync™
Grades 6–8
Problem-Solving Strategies

Answers

Part 1

Step 1: Understand the problem.
Step 2: Develop a plan.
Step 3: Solve the problem.
Step 4: Look back.

Part 2

1. a. Jennifer goes swimming every three days. She rides her bike every four days.
   b. How many times in 30 days will she do both activities?
2. Student responses may include: Make a Table
3. She will do both activities on two days, days 12 and 24.
4. Student responses may vary.

Part 3

1. Student responses may include: Draw a Diagram. Mrs. Ellison will need 30 tiles to cover her floor.
2. Student responses may include: Make a List or Draw a Tree Diagram. Mitchell will be able to wear 48 different outfits.
3. Student responses may include: Work Backwards. Alyssa started with 18 bracelets.
4. Student responses may include: Write an Equation or Guess and Test. It will take Milo three hours to complete his assignment.

Mathematics for SylvanSync™
Grades 6–8
Pythagorean Theorem

Answers
Part 1
1. The hypotenuse is the longest side. The other two sides are the legs.
2. Hypotenuse: 5 units; Legs: 3 units, 4 units
3. White (hypotenuse): 25 squares; green (leg): 16 squares; blue (leg): 9 squares
4. 16 + 9 = 25
5. Student responses may vary. Acceptable answers include: The number of squares on the hypotenuse is equal to the sum of the squares on the legs.
6. White (hypotenuse): 5^2; green (leg): 4^2; blue (leg): 3^2; $3^2 + 4^2 = 5^2$

Part 2
1. Check that students have drawn the triangle correctly.
2. Check student's work.
3. 25 squares; 144 squares; 169 squares
5. 25 + 144 = 169
6. $5^2 + 12^2 = 13^2$
7. Yes, the relationship is the same.

Mathematics for SylvanSync™
Grades 6–8
Pascal's Triangle

Answers

1. 1, 9, 36, 84, 126, 126, 84, 36, 9, 1

   1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1

2. The first five numbers are repeated in reverse order.

3. The numbers in the second diagonal are the whole numbers, or counting numbers.

4. The differences between the numbers in the third diagonal are the counting numbers:
   1 + **2** = 3; 3 + **3** = 6; 6 + **4** = 10; 10 + **5** = 15; 15 + **6** = 21

5. A triangular number is the number of dots in an equilateral triangle evenly filled with dots. For example, three dots, 6 dots, 10 dots, and 15 dots can each be arranged in a triangle and so they are triangular numbers.

6. Row1: 1
   Row 2: $1 + 1 = \mathbf{2}$
   Row 3: $1 + 2 + 1 = \mathbf{4}$
   Row 4: $1 + 3 + 3 + 1 = \mathbf{8}$
   Row 5: $1 + 4 + 6 + 4 + 1 = \mathbf{16}$
   Row 6: $1 + 5 + 10 + 10 + 5 + 1 = \mathbf{32}$
   The sum of each row is twice the sum of the previous row.

7. Row 7: $1 + 6 + 15 + 20 + 15 + 6 + 1 = \mathbf{64}$

8. The numbers on either side of the pencil repeat from the pencil out towards the edges.

9. Student responses may vary. Check student’s work.

Optional Activity

1. Student responses may vary. Check student’s work.
2. Student responses may vary. Check student’s work.
3. Student responses may vary. Check student’s work.

Mathematics for SylvanSync™
Grades 6–8
Exploring Integers

Answers
Part 1
1. a. 6 yellow counters
   b. 4 red counters
   c. 11 yellow counters
   d. 9 red counters

2. a. –2
   b. +5, or 5
   c. –10

Part 2
1. a. 4 + (–10) = –6
   b. 3 + 6 = 9
   c. –7 + (–5) = –12
   d. –2+ 9 = 7

2. Suggested: The sum has the same sign as the addend with the greater absolute value. The sign is the same as the larger pile of counters.

Part 3
8. a. 4 + (–7) = –3
   b. 7 + (–4) = 3
   c. 12 + 3 = 15
   d. –3 + (–12) = –15
   e. –8 + (–10) = –18
   f. 10 + 8 = 18
   g. –6 + 5 = –1
   h. –5 + 6 = 1

9. If you change the order of the numbers, the difference has the opposite sign.

Mathematics for SylvanSync™
Grades 6–8
Probability Activities

Answers
Part 1

1. Student responses may vary.
2. The first part of the tree shows the numbers 1 through 6. Each of those numbers has six branches, showing the numbers 1 through 6. The possible outcomes are shown below:

   (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)

   (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)

   (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)

   (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)

   (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)

   (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)

3. $\frac{1}{6}$

   The favorable outcomes are (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), and (6, 6).

   $$P = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{6}{36} = \frac{1}{6}$$

4. $\frac{1}{4}$

   The favorable outcomes are (2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), and (6, 6).

   $$P = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{9}{36} = \frac{1}{4}$$

Mathematics for SylvanSync™
Grades 6–8
Probability Activities

Part 2

1. $\frac{1}{4}$, 0.25, or 25%
2. $\frac{3}{4}$, 0.75, or 75%
3. $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$
4. $\frac{1}{4} \times \frac{2}{4} = \frac{2}{16} = \frac{1}{8}$

Part 3

1. Student responses may vary.
2. $\frac{1}{6}$
3. Student responses may include: Differences are due to the size of the sample, because experimental probability approaches theoretical probability as the sample size increases.

Mathematics for SylvanSync™
Grades 6–8
Geometric Solids

Answers
Part 1
1. shapes that slide: cylinder, square pyramid, cube, hexagonal prism, octagonal prism, triangular prism, rectangular prism, cone, hemisphere

   shapes that roll: sphere, cylinder, cone, hemisphere

   shapes that do both: cylinder, cone, hemisphere

2. Student responses may vary. Acceptable answers include: sort by solids that can stand on more than one face, one face, or cannot stand at all.

Part 2
1. Check that the student has correctly identified the shapes of the faces of the solid.
2. Check that the student has assembled the solid correctly.
3. Check that the student has correctly identified the shapes of the faces of the solid and correctly assembled the solid.

Part 3
1. Check that the student has correctly identified the square pyramid.
2. Check that the student has correctly identified the cube.
3. Check that the student has correctly identified the hexagonal prism.
4. Check that the student has correctly identified three additional prisms, including the triangular prism, the square prism, and the rectangular prism.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Answers
Part 1

1. Check that students have chosen the correct Fraction Tower Cubes.

   a. $\frac{25}{100}$

   b. $\frac{10}{100}$

   c. $\frac{20}{100}$

   d. $\frac{50}{100}$

2. Check that students have chosen the correct Percent Tower Cubes.

   a. 25%

   b. 10%

   c. 20%

   d. 50%

   For each fraction, the number on the Percent Tower Cubes is equal to the numerator of the answer fraction in Question 1.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Part 2

1. Check that students have chosen the correct Fraction Tower Cubes.

   a. $\frac{75}{100}$

   b. $\frac{40}{100}$

   c. $\frac{70}{100}$

2. Check that students have chosen the correct Percent Tower Cubes.

   a. 75%

   b. 40%

   c. 70%

   For each fraction, the sum of the numbers on the Percent Tower Cubes is equal to the numerator of the answer fraction in Question 1.

3. Student responses may vary. Acceptable answers include: Since a fraction with a denominator of 100 divides a whole into 100 parts, the number of parts of the fraction will be the same as the equivalent percent, which is also the number of parts out of 100.

Mathematics for SylvanSync™
Grades 6–8
Understanding Percent

Part 3

1. Check that students have chosen the correct Percent Tower Cubes.

   a. 1 × 50%; 2 × 25%; 3 × 16.6%; 4 × 12.5%; 5 × 10%; 6 × 8.3%

   b. 1 × 25%; 2 × 12.5%; 3 × 8.3%

   c. 2 × 20%; 4 × 10%

   d. 1× 33.3%; 2 × 16.6%; 4 × 8.3%

2. Check that students have chosen the correct Fraction Tower Cubes.

   a. $\frac{1}{2}; \frac{2}{4}; \frac{3}{6}; \frac{4}{8}; \frac{5}{10}; \frac{6}{12}$

   b. $\frac{1}{4}; \frac{2}{8}; \frac{3}{12}$

   c. $\frac{2}{5}; \frac{4}{10}$

   d. $\frac{1}{3}; \frac{2}{6}; \frac{4}{12}$

Mathematics for SylvanSync™
Grades 9-12
Absolute Value

Answers
Part 1
1. Check student's table.
2. Check student's work.
3. The last two columns for each set of numbers are equal.

Part 2
1. Check student's table.
2. Check student's work.
3. The last two columns for each set of numbers are not equal.

Part 3
1. Check student's table.
2. Check student's work.
3. The last two columns for each set of numbers are equal.

Mathematics for SylvanSync™
Grades 9-12
Matrices

Answers
Part 1

1. $A=\begin{bmatrix} 19 & 47 & 21 \\ 30 & 47 & 25 \end{bmatrix}$, columns may be in a different order, and rows may be switched. However, since the directions specified a 2 x 3 matrix and not a 3 x 2 matrix it is important that there are exactly two rows and three columns.

2. Each row represents the type of driving—either city or highway.

3. Each column represents a different type of vehicle: sports car, hybrid sedan, or truck.

4. $1.10A=\begin{bmatrix} 20.9 & 51.7 & 23.1 \\ 33 & 51.7 & 27.5 \end{bmatrix}$

5. The matrix $1.10A$ represents the result if each of the miles per gallon rates were increased by 10%.

Part 2

1. The matrices must have the same dimensions.

2. $M+N=\begin{bmatrix} -1 & -6.25 \\ 20 & 11\frac{1}{2} \\ \frac{1}{10} & 3 \\ -6 & 6.2 \end{bmatrix}$

3. $N+M=\begin{bmatrix} -1 & -6.25 \\ 20 & 11\frac{1}{2} \\ \frac{1}{10} & 3 \\ -6 & 6.2 \end{bmatrix}$

Mathematics for SylvanSync™
Grades 9-12
Matrices

4. No; the commutative property

5. No. $M - N = \begin{bmatrix} -5 & 7.75 \\ -2 & 12\frac{1}{2} \\ \frac{7}{10} & -13 \\ 6 & -4.2 \end{bmatrix}$ $N - M = \begin{bmatrix} 5 & -7.75 \\ 2 & -12\frac{1}{2} \\ -\frac{7}{10} & 13 \\ -6 & 4.2 \end{bmatrix}$

Mathematics for SylvanSync™
Grades 9-12
Equation of a Line

Answers
Part 1

1. $y = \frac{1}{2}x - 1$
2. $x - 2y = 2$
3. $y = \frac{3}{10}x + 11$
4. $3x - 10y = -110$

Part 2

1. $3x + 4y = -28$
2. $y = -\frac{5}{6}x + \frac{10}{3}$
3. $y = 2$
4. $3x - y = 2$

Mathematics for SylvanSync™
Grades 9-12
Polynomials

Answers
Part 1

1. Quartic Binomial
2. Constant Monomial
3. Cubic Trinomial
4. Quadratic Binomial

Part 2

1. $4x^2 - 3x - 13$
2. $4x^2 - 8xy - 3y$
3. $7d^3 - 4a^2 + 5a - 6$
4. $-4n^3 + 10n^2 - 5n - 8$

Mathematics for SylvanSync™
Grades 9-12
Multiply Polynomials

Answers

1. $4x^3 + x^2 + 7x + 15$
2. $3x^2 + 2xy + 5x - 8y^2 + 10y$
3. $-30m^4 + 15m^2 - 10$
4. $6x^2yz - 42x^2yz + 12xyz^2$
5. $x^4 - x^3 - 14x^2 + 22x - 8$
6. $3a^2 + 16ab - 15ac - 5b^2 + 33bc - 18c^2$

Mathematics for SylvanSync™
Grades 9-12
Multiplying Binomials

Answers
Part 1
1. $x^2 - 4$
2. $x^2 - 36$
3. $4x^2 - 9$
4. $x^2 - 49y^2$
5. $y^2 - 25$
6. The binomials have the same terms, but opposite signs. Each answer has 2 terms. All of the coefficients and constants are perfect squares.
7. $m^2 - n^2$

Part 2
1. $x^2 + 10x + 25$
2. $a^2 - 8a + 16$
3. $4x^2 - 12x + 9$
4. $9y^2 + 24y + 16$
5. $25x^2 - 30xy + 9y^2$
6. The binomials are being squared. Each answer has 3 terms.
7. $m^2 - 2mn + n^2$and $m^2 + 2mn + n^2$

Part 3
1. $x^2 - 2x - 24$
2. $3x^2 + 17x + 10$
3. $4a^2 - 25a - 21$
4. $2x^2 - 21x + 40$
5. $ac - ad - bc + bd$
6. $ac + ad + bc + bd$
7. $ac + ad - bc - bd$

Mathematics for SylvanSync™
Grades 9-12
Factoring Binomials and Trinomials

Answers
Part 1
1. $5(x-3)$
2. $yz(z^2-3z+2)$
3. $6x(x^2-4x+1)$
4. $3ab^2(ab+3-4a)$

Part 2
1. $(x-10)(x+10)$
2. $(3a-7)(3a+7)$
3. $(4x-y)(4x+y)$
4. Not Factorable

Part 3
1. $(x-3)^2$
2. $(m+4)^2$
3. $(2y+5)^2$
4. $(4b-1)^2$

Part 4
1. $x(x-2)(x+2)$
2. $6(a-3)(a+3)$
3. $2(y-5)^2$
4. $8(a^2-ab+b^2)$

Mathematics for SylvanSync™
Grades 9-12
Polynomial Division

Answers
Part 1
1. $x^2 - 4x + 7$
2. $2x + \frac{5}{4} + \frac{3}{4x}$
3. $-\frac{x^2}{2} + 6x + 3 + \frac{2}{x}$
4. $2x^2 - \frac{2x}{3} + \frac{7}{3}$
5. $x^2 - 2x + \frac{5}{7} - \frac{1}{7x}$

Part 2
1. $x^3 - 5x^2 - 6x - 5 - \frac{19}{x-2}$
2. $x - 1 + \frac{2}{x-1}$
3. $x^2 - x + 1$
4. $3x - 7$
5. $3x^2 - 4x - 2 + \frac{1}{x-1}$

Mathematics for SylvanSync™
Grades 9-12
Pythagorean Theorem Model

Answers
Part 1
1. Check that students have drawn the triangle correctly.

2. Check student's work.

3. 64 squares; 225 squares; 289 squares

4. 64 + 225 = 289

5. $8^2 + 15^2 = 17^2$

6. Yes, by substituting the smallest sides, 8 and 15, for the length of the legs, and 17 for the hypotenuse, the equation $8^2 + 15^2 = 17^2$ is true which represents the Pythagorean Theorem.

Part 2
1. Check that students have the triangle put together correctly.

2. The result is 13 = 16. No, it is not a true statement because 13 does not equal 16.

3. Student should find one of the angles to be obtuse, so it is an obtuse triangle.

Part 3
1. Check that students have the triangle put together correctly.

2. The result is 18 = 16. No, it is not a true statement because 18 does not equal 16.

3. Student should find all of the angles to be acute, so it is an acute triangle.

Part 4
1. Obtuse triangle, check student's work.

2. Acute triangle, check student's work.

Mathematics for SylvanSync™
Grades 9-12
Similar Polygons

Answers
Part 1
1. Check student’s work to make sure the corresponding vertices match.
2. Check student’s work to make sure corresponding angles are congruent.
3. Check student’s work to make sure corresponding sides are proportional.
4. Check student’s work. Yes, the ratios should be the same.

Part 2
1. Check student’s work.
2. Check student’s work to make sure corresponding sides are in the ratio of 2 to 3.

Part 3
1. Check student’s work.
2. Check student’s work to make sure corresponding sides are proportional.
3. Check student’s work.